

MATH 8090: An Overview of Time Series Analysis

Readings: CC08 Chapter 1; SS17 Chapter 1.1; BD16 Chapter
1.1–1.3

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Week 1

Agenda

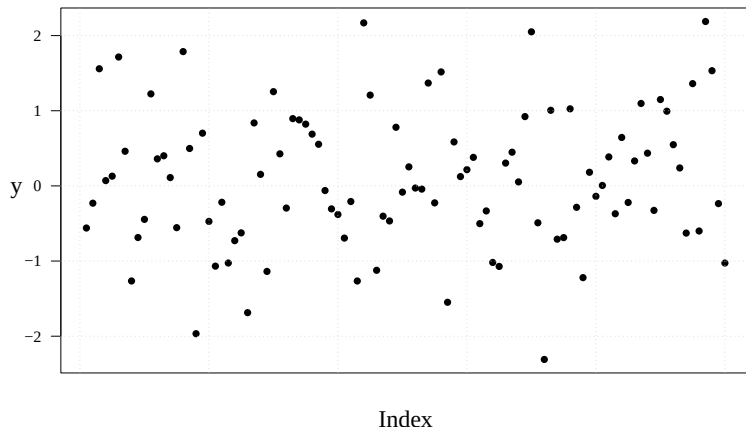
Brief Review of Statistical Inference

Time Series Data

Time Series Models

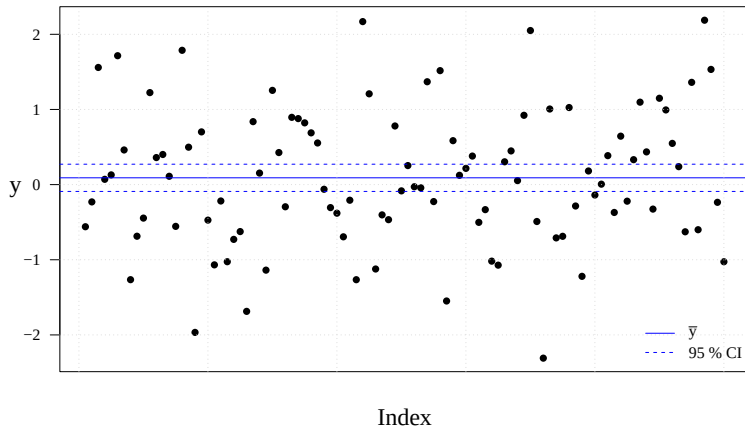
Objectives of Time Series Analysis

Statistical Inference from a Sample



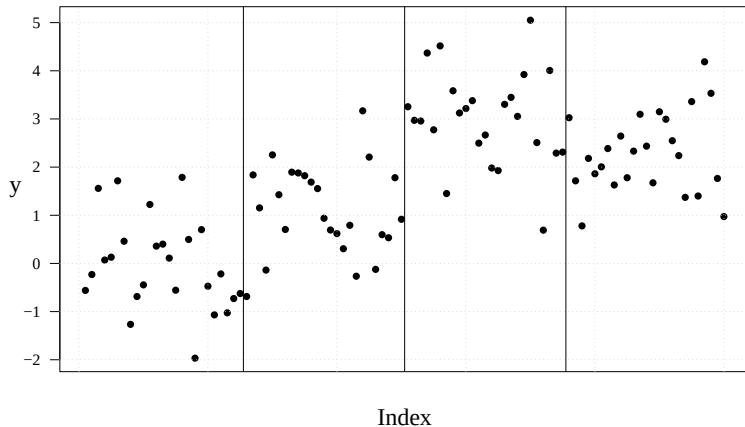
How can we use a random sample to infer the population mean?

Statistical Inference from a Sample



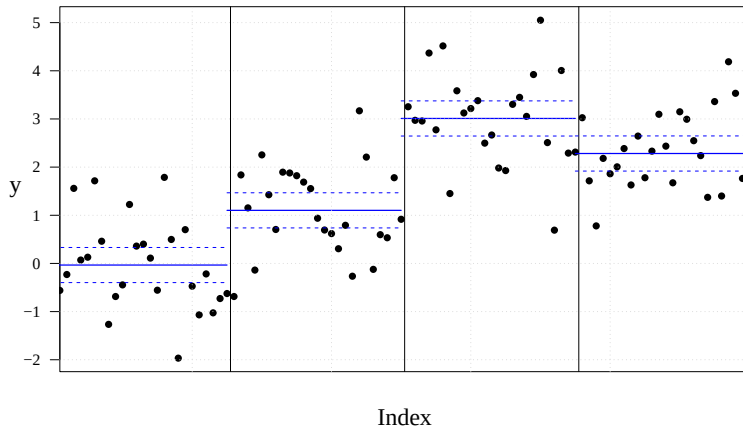
How do we obtain the point and interval estimates?

Statistical Inference for Group Means



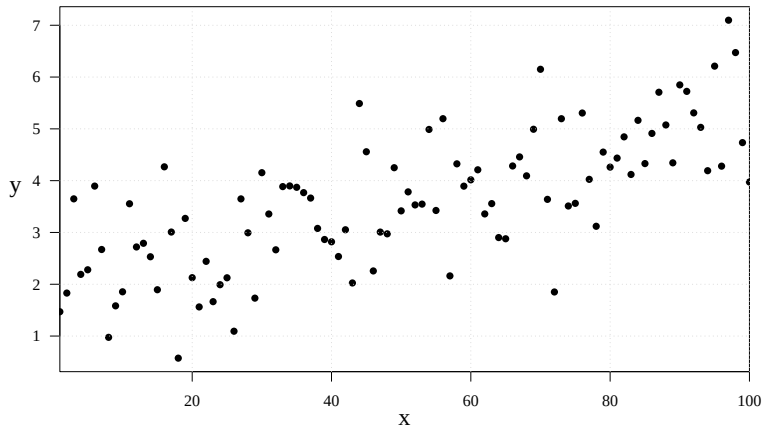
Are the group means significantly different?

Statistical Inference for Group Means



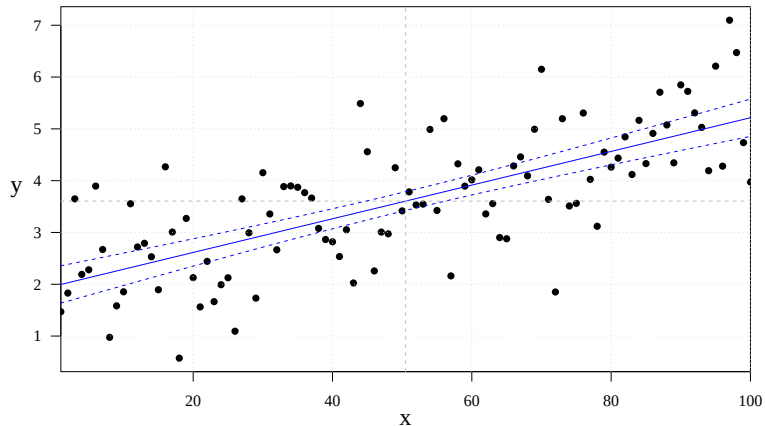
Which statistical technique is used here?

Statistical Inference for Conditional Mean



How does y change conditionally on x ?

Statistical Inference for Conditional Mean Curve

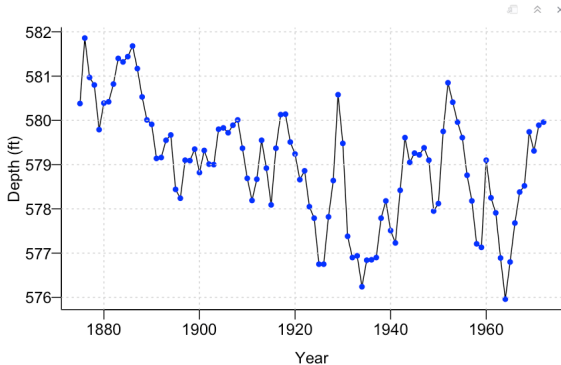


Which statistical technique is used here?

Level of Lake Huron 1875–1972

Annual measurements of the level of Lake Huron in feet. [Source: Brockwell & Davis, 1991]

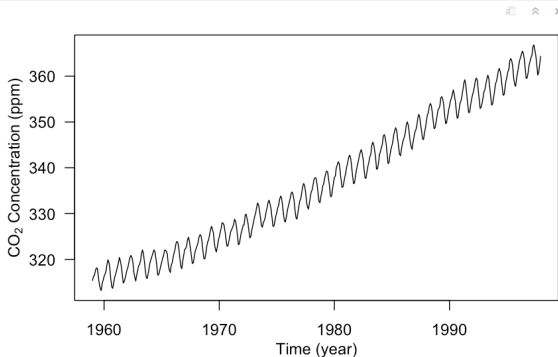
```
## {r}  
par(mar = c(3.2, 3.2, 0.5, 0.5), mgp = c(2, 0.5, 0), bty = "L")  
data(LakeHuron)  
plot(LakeHuron, ylab = "Depth (ft)", xlab = "Year", las = 1)  
points(LakeHuron, cex = 0.8, col = "blue", pch = 16)  
grid()  
##
```



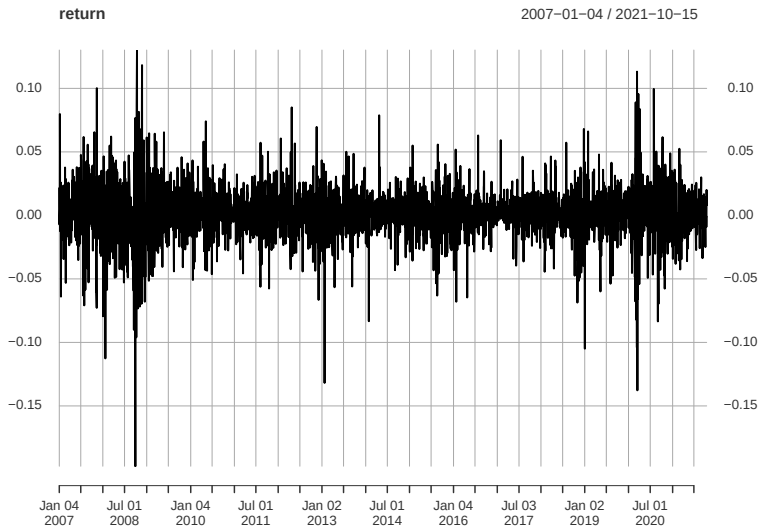
Mauna Loa Atmospheric CO₂ Concentration

Monthly atmospheric concentrations of CO₂ at the Mauna Loa Observatory [Source: [Keeling & Whorf, Scripps Institution of Oceanography](#)]

```
##{r}
data(co2)
par(mar = c(3.8, 4, 0.8, 0.6))
plot(co2, las = 1, xlab = "", ylab = "")
mtext("Time (year)", side = 1, line = 2)
mtext(expression(paste("CO"[2], " Concentration (ppm)")), side = 2, line = 2.5)
##
```

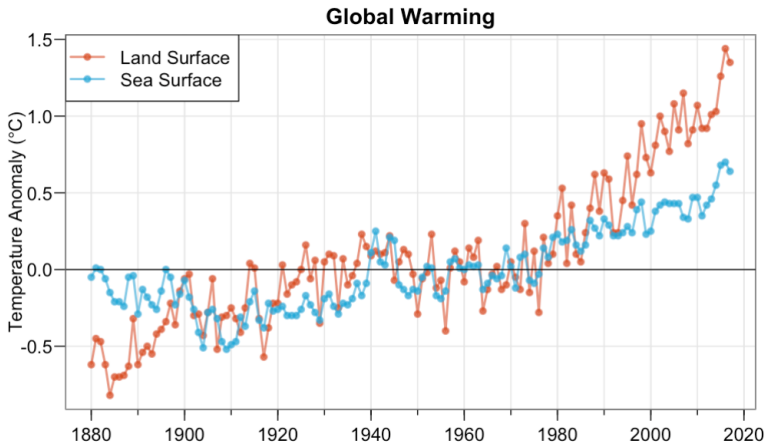


Apple Stock Log Returns



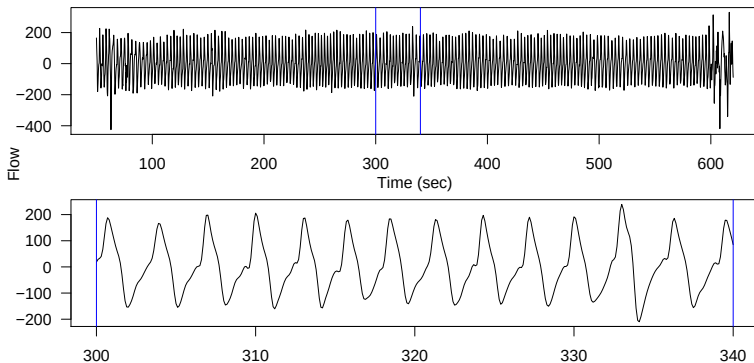
Global Annual Temperature Anomalies

[Source: [NASA GISS Surface Temperature Analysis](#)]



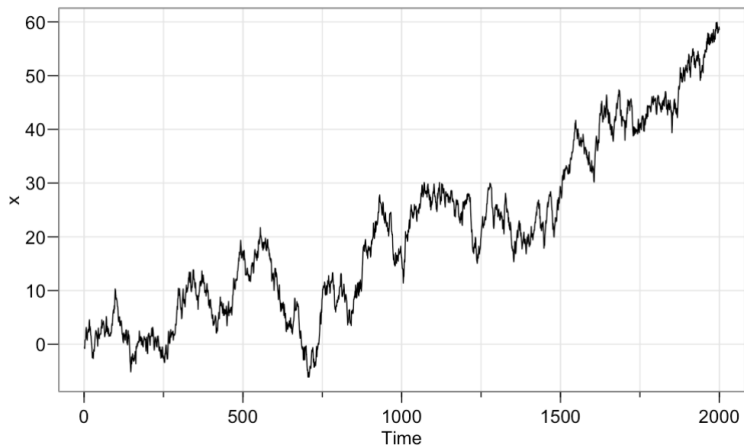
Sleep Airflow Signal

A “normal” patient’s 10 Hz sleep airflow signal [Source: [H. et al. 2022](#)]



A Simulated Time Series

```
## {r}  
set.seed(123)  
w <- rnorm(2000); x <- cumsum(w); tsplot(x, las = 1)
```



Time Series Data

- ▶ A **time series** is a collection of observations $\{y_t, t \in T\}$ taken sequentially in time (t) with the index set T
 - ▶ $T = \{0, 1, 2, \dots, T\} \subset \mathbb{Z} \Rightarrow$ **discrete-time time series**
 - ▶ $T = [0, T] \subset \mathbb{R} \Rightarrow$ **continuous-time time series**
- ▶ A discrete-time time series might be intrinsically discrete or might arise from an underlying continuous-time time series via
 - ▶ sampling (e.g., instantaneous wind speed)
 - ▶ aggregation (e.g., daily accumulated precipitation amount)
 - ▶ extrema (e.g., daily maximum temperature)
- ▶ We will focus on **discrete-time real-valued** ($Y_t \in \mathbb{R}$) **time series**

Exploratory Time Series Analysis

- ▶ Start with a **time series plot**, i.e., to plot y_t versus t 
- ▶ Look for:
 - ▶ Are there abrupt changes?
 - ▶ Are there “outliers”?
 - ▶ Is there a need to transform the data?
- ▶ Examine the **trend**, **seasonal components**, and the “noise” term

Features of Time Series

► Trends

- A trend, μ_t , represents smooth, long-term changes, typically in the mean $\Rightarrow$ *“the essential idea of trend is that it shall be smooth”* - [Kendall, 1973]
- Usually the form of the trend is unknown and needs to be estimated. When the trend is removed, we obtain a **detrended** series

► Seasonal or periodic components

- A seasonal component s_t repeats itself in time, i.e., $s_t = s_{t+kd}$
- We need to estimate the form and/or the period d of the seasonal component to **deseasonalize** the series

► The “noise” process

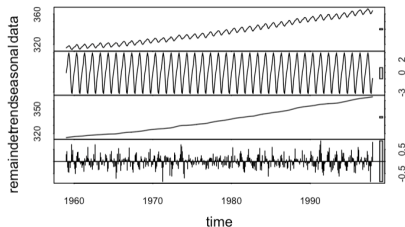
- The noise process, η_t , is the component that is neither trend nor seasonality
- We will focus on finding plausible (typically stationary) statistical models for this process

Combining Trend, Seasonality, and Noise Together

Two common decompositions are:

- Additive model:

$$y_t = \mu_t + s_t + \eta_t, \quad t = 1, \dots, T$$



- Multiplicative model:

$$y_t = \mu_t s_t \eta_t, \quad t = 1, \dots, T$$

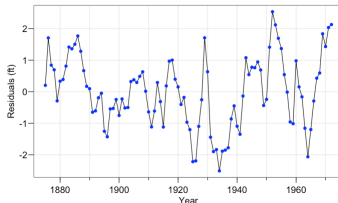
If all $\{y_t\}$ are positive then we obtain the additive model by taking logarithms:

$$\log y_t = \log \mu_t + \log s_t + \log \eta_t, \quad t = 1, \dots, T$$

Time Series Models

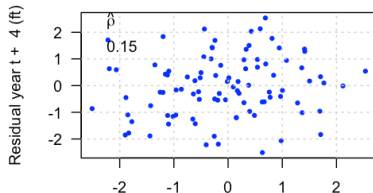
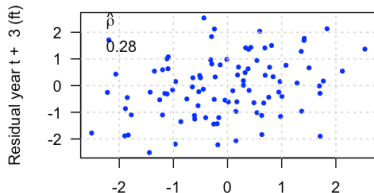
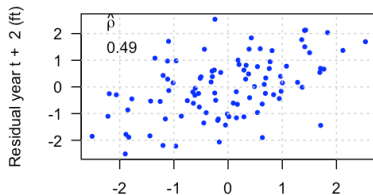
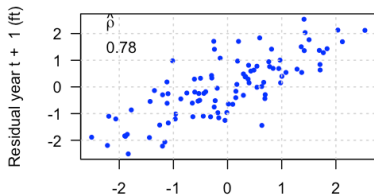
Lake Huron Time Series

- ▶ **Time series analysis** studies observations that are **dependent** across time (typically $\{\eta_t\}$)
- ▶ Some key features of the Lake Huron time series:
 - ▶ decreasing trend
 - ▶ some “random” fluctuations around the decreasing trend
- ▶ We extract the “noise” component by assuming a linear trend



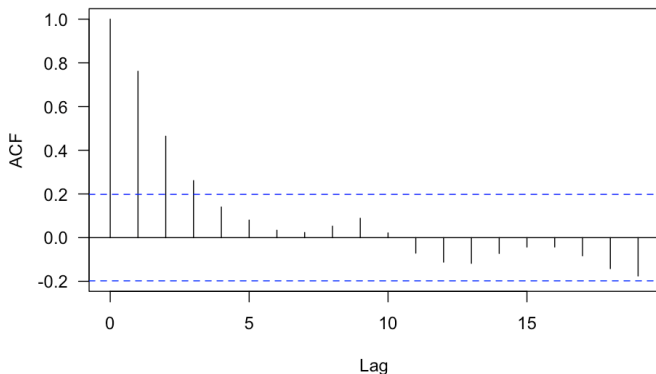
Exploring the Temporal Dependence Structure of $\{\eta_t\}$

Nearby observations tend to be more similar than observations farther apart in time. We can explore this using lag plots



Further Exploration of the Temporal Dependence Structure

Let's plot the correlation as a function of the time lag



In a few weeks we will learn how to use this information to suggest an appropriate model

Time Series Models

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- ▶ A central simplifying assumption in time-series modeling is **stationarity** $\Rightarrow$ distributional properties depend only on time lags, not on specific time points
- ▶ Most time series are not stationary; we remove or model non-stationarity (e.g., detrending, deseasonalizing) to obtain a stationary component $\{\eta_t\}$. Typically, we assume **second-order stationarity**:

$$\begin{aligned} \mathbb{E}[\eta_t] &= 0 \text{ for all } t, \text{ and} \\ \text{Cov}(\eta_t, \eta_{t'}) &= \gamma(t' - t) = \text{Cov}(\eta_{t+s}, \eta_{t'+s}) \end{aligned}$$

Objectives of Time Series Analysis

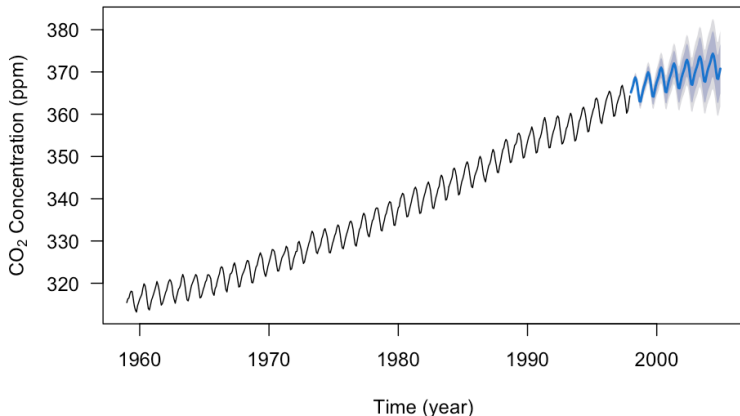
Objectives of Time Series Analysis: Modeling

- ▶ **Modeling:** find a **statistical model** that adequately explains the observed time series
- ▶ For example, identify a model that accounts for the fact that Lake Huron depths are correlated across different years and exhibit a decreasing long-term trend
- ▶ The fitted model can be used for further **statistical inference**, for instance, to answer questions such as: **Is there evidence of a decreasing trend in Lake Huron depths?**

Objectives of Time Series Analysis: Forecasting

Forecasting is perhaps the most common objective. One observes a time series of given length and wishes to **predict** or **forecast** future values based on the observed data

Forecasts from TBATS(1, {3,1}, -, {<12,5>})



Other Objectives of Time Series Analysis

- ▶ **Adjustment:** e.g., [seasonal adjustment](#), where the seasonal component is estimated and removed to better reveal the underlying trend
- ▶ **Simulation:** use a time series model (that adequately describes a physical process) as a surrogate to *simulate repeatedly and approximate the process's behavior*
- ▶ **Control:** adjust [input \(control\)](#) parameters to make the series conform more closely to a given standard (many examples come from statistical quality control)

Time Series in the Era of AI

- ▶ Time-series forecasting has long been a central problem in statistics:

past observations $\longrightarrow$ future prediction.

- ▶ Classical statistical approaches:
 - ▶ ARIMA and exponential smoothing
 - ▶ State-space models
 - ▶ GARCH and other dynamic models
- ▶ Modern ML/AI approaches:
 - ▶ Tree-based and ensemble methods
 - ▶ Recurrent neural networks (RNN/LSTM)
 - ▶ Transformers
 - ▶ Time-series foundation models

A common goal

Learn temporal structure from observed data to **understand, predict, and quantify future behavior.**

What Does AI Change?

Traditional Time-Series Models

- ▶ Explicit model structure
- ▶ Carefully specified dependence
- ▶ Relatively few parameters
- ▶ Statistical interpretability
- ▶ Uncertainty quantification

Modern ML / AI Models

- ▶ Flexible nonlinear structure
- ▶ Learn complex temporal patterns
- ▶ Potentially many parameters
- ▶ Benefit from large datasets

Not “Statistics vs. AI”

Modern forecasting increasingly combines **statistical structure, machine learning, and domain knowledge.**

A New Direction: Time-Series Foundation Models

Traditional approach

one time series $\longrightarrow$ fit one model $\longrightarrow$ forecast

Foundation-model approach

many time series $\longrightarrow$ pretrained model $\longrightarrow$ forecast a new series

- ▶ Learn **general temporal patterns** that transfer across datasets
- ▶ Enable ideas such as **zero-shot** and **probabilistic forecasting**

Key idea

Can we forecast a new time series without fitting a model from scratch?

Why Learn Classical Time Series in the AI Era?

- ▶ What does **temporal dependence** mean?
- ▶ What are **stationarity, autocorrelation, and seasonality**?
- ▶ How do we distinguish **signal, dependence, and noise**?
- ▶ How should forecasts be **evaluated**?
- ▶ How do we quantify **forecast uncertainty**?
- ▶ When does a model fail because the data-generating process changes?

Throughout this course

The statistical foundations we develop will help us understand **both classical time-series models and modern AI forecasting methods**